

Effect Estimation Using Model-Based Recursive Partitioning and Bayesian Flexible Model

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ABSTRACT

Personalized treatment strategies are central to precision medicine, yet treatment effects often vary substantially across patient subpopulations. Conventional approaches that focus on average treatment effects may obscure clinically meaningful heterogeneity and lead to suboptimal decision-making. This study proposes an integrated framework for personalized treatment effect estimation that combines model-based recursive partitioning with Bayesian flexible models. Model-based recursive partitioning is employed to identify covariate-driven subgroups exhibiting parameter instability, enabling data-adaptive discovery of clinically interpretable patient strata. Within each identified subgroup, Bayesian flexible models are used to estimate treatment effects while capturing complex, nonlinear relationships and providing full posterior uncertainty quantification. The proposed approach facilitates robust estimation of conditional and individual treatment effects, accommodates high-dimensional covariates, and allows partial information sharing across subgroups through hierarchical priors. Simulation studies demonstrate improved estimation accuracy, coverage probabilities, and subgroup-specific interpretability compared with conventional recursive partitioning and non-Bayesian methods. An empirical application to real-world healthcare data illustrates the clinical relevance of the identified subgroups and highlights the advantages of Bayesian inference for uncertainty-aware personalized treatment recommendations. Overall, this framework offers a principled and flexible solution for heterogeneous treatment effect estimation in personalized medicine.

Keywords: Personalized treatment effects, Heterogeneous treatment effects, Model-based recursive partitioning, Bayesian flexible models, Causal inference, Precision medicine

1. INTRODUCTION

Personalized medicine aims to tailor treatment strategies to individual patient characteristics to maximize clinical benefits and minimize adverse effects. Traditional approaches often focus on estimating the average treatment effect (ATE) across a population, which can obscure substantial

variation in treatment response among subgroups. Recognizing and quantifying heterogeneous treatment effects (HTEs) is therefore critical for effective clinical decision-making and improving patient outcomes.

Model-based recursive partitioning (MOB) offers a systematic approach for subgroup discovery by identifying covariate-driven splits where treatment effects vary. However, MOB alone may struggle to capture complex, nonlinear relationships within subgroups, especially in high-dimensional settings. Bayesian flexible models, such as Bayesian additive regression trees (BART), Gaussian processes, or hierarchical splines, provide a powerful framework for modeling complex interactions, estimating uncertainty, and enabling partial information sharing across subgroups.

This study integrates model-based recursive partitioning with Bayesian flexible modeling to provide a robust and interpretable framework for personalized treatment effect estimation. By first identifying subgroups where treatment effects differ and then applying Bayesian models to estimate these effects within each subgroup, the framework captures both heterogeneity and uncertainty. The proposed approach aims to improve estimation accuracy, enhance interpretability of subgroup-specific effects, and support more informed clinical decision-making.

Objectives of the study:

1. To develop an integrated framework combining model-based recursive partitioning and Bayesian flexible modeling for estimating personalized treatment effects.
2. To assess the framework's performance in identifying subgroups with heterogeneous treatment effects.
3. To demonstrate its applicability using real-world healthcare data.

2. LITERATURE REVIEW

2.1 Heterogeneous Treatment Effects in Personalized Medicine

Heterogeneous treatment effects (HTEs) occur when the impact of a treatment varies across patients due to differences in baseline characteristics, genetics, comorbidities, or other covariates. Estimating HTEs is essential in personalized medicine, as average effects may be misleading. Several methods have been proposed, including stratified analyses, propensity score-based approaches, and machine learning techniques such as causal forests and Bayesian models.

2.2 Model-Based Recursive Partitioning (MOB)

Model-based recursive partitioning is a tree-based approach that identifies subgroups by detecting parameter instability in a given model. MOB splits the data based on covariates that significantly modify treatment effects, providing an interpretable structure of subgroups. Advantages include the ability to handle high-dimensional covariates and to offer visual representation of subgroup effects. Limitations include potential overfitting and sensitivity to small sample sizes within terminal nodes.

2.3 Bayesian Flexible Models

Bayesian flexible models provide a probabilistic framework for modeling complex, nonlinear relationships. Methods such as Bayesian additive regression trees (BART), Gaussian processes, and Bayesian splines allow for flexible function estimation, hierarchical modeling, and uncertainty quantification. Bayesian methods also facilitate partial pooling across subgroups, improving estimation stability, particularly in small-sample or high-dimensional settings.

2.4 Integration of MOB and Bayesian Models

Recent studies suggest combining tree-based methods with Bayesian modeling to leverage the strengths of both approaches. Recursive partitioning identifies interpretable subgroups with differential treatment effects, while Bayesian flexible models refine effect estimates within these subgroups, capturing nonlinear patterns and quantifying uncertainty. Despite promising results, limited research has systematically integrated these approaches in the context of personalized medicine, leaving opportunities for methodological advancement.

2.5 Gaps and Research Motivation

While existing approaches can estimate HTEs, many fail to simultaneously achieve interpretability, robustness, and uncertainty quantification. The integration of model-based recursive partitioning with Bayesian flexible models addresses these gaps by combining subgroup discovery, flexible modeling, and probabilistic inference, making it a compelling strategy for personalized treatment effect estimation.

3. METHODOLOGY

3.1 Conceptual Framework and Causal Assumptions

The proposed framework relies on the **potential outcomes framework**, where for each individual i , $Y_i(1)$ and $Y_i(0)$ denote potential outcomes under treatment and control, respectively. The **individual treatment effect (ITE)** is defined as:

$$\tau_i = Y_i(1) - Y_i(0)$$

We aim to estimate **conditional average treatment effects (CATE)** given covariates X_i :

$$\tau(X_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i]$$

Causal assumptions:

1. **Unconfoundedness:** $\{Y_i(1), Y_i(0)\} \perp T_i \mid X_i$
2. **Overlap:** $0 < P(T_i = 1 \mid X_i) < 1$
3. **Stable Unit Treatment Value Assumption (SUTVA)**

These assumptions ensure that treatment effect estimates are unbiased and identifiable from observed data.

3.2 Model-Based Recursive Partitioning (MOB) for Subgroup Discovery

MOB identifies subgroups with heterogeneous treatment effects by detecting **parameter instability** with respect to covariates:

1. Fit a **global model**: e.g., linear or logistic regression
2. Test for **parameter instability** across covariates using statistical tests (e.g., score or likelihood-ratio tests)
3. Recursively split the dataset along covariates with significant instability
4. Stop splitting based on **predefined criteria**: minimum node size, p-value threshold, or cross-validation

The result is an interpretable tree structure, where each terminal node represents a subgroup with potentially distinct treatment effects.

3.3 Bayesian Flexible Models Within Subgroups

Within each identified subgroup, we apply a **Bayesian flexible model** to estimate treatment effects:

- **Model form:**

$$Y_i \mid X_i, T_i \sim f(Y_i \mid T_i, X_i, \theta)$$

- **Priors:** Hierarchical priors allow partial pooling across subgroups:

$$\theta_j \sim \text{Normal}(\mu_\theta, \sigma_\theta^2), j = 1, \dots, J$$

- **Posterior inference:** Using Markov Chain Monte Carlo (MCMC) or Hamiltonian Monte Carlo to estimate posterior distributions of treatment effects, enabling uncertainty quantification with credible intervals.

Bayesian flexible models capture nonlinear relationships between covariates and outcomes, improving estimation accuracy in complex, high-dimensional settings.

3.4 Integration of MOB and Bayesian Models

Two integration strategies are considered:

1. Sequential approach:

- Step 1: Use MOB to identify subgroups
- Step 2: Fit Bayesian models separately within each subgroup

2. Joint approach (optional extension):

- Bayesian hierarchical model incorporates partitioning within the prior structure, allowing simultaneous subgroup discovery and effect estimation

Sequential integration is computationally simpler, while joint modeling enables full uncertainty propagation.

3.5 Estimation of Personalized Treatment Effects

1. **Conditional Average Treatment Effects (CATE):** Posterior mean of treatment effect within each subgroup
2. **Individual Treatment Effects (ITE):** Posterior predictive distribution for each individual
3. **Uncertainty quantification:** 95% credible intervals to reflect estimation uncertainty

Visualization techniques (tree plots, posterior density plots, and interaction maps) help interpret heterogeneous effects.

4. SIMULATION STUDY

4.1 Simulation Design

Synthetic datasets are generated to evaluate performance under controlled heterogeneous treatment effect scenarios:

1. **Covariates:** 20-50 continuous and categorical variables
2. **Treatment assignment:** Randomized or covariate-dependent
3. **Outcome model:**

$$Y_i = f(X_i) + \tau(X_i)T_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

where $\tau(X_i)$ varies across subgroups, including nonlinear interactions.

4.2 Performance Metrics

- **Bias:** Difference between estimated and true treatment effects
- **Root Mean Squared Error (RMSE):** Accuracy of subgroup and individual effect estimates
- **Coverage probability:** Proportion of true effects within 95% credible intervals
- **Subgroup identification accuracy:** Proportion of correctly identified splits

4.3 Comparative Methods

The proposed framework is compared against:

- Standard MOB without Bayesian modeling
- Bayesian additive regression trees (BART) applied to the full dataset
- Causal forests

4.4 Sensitivity Analysis

- Varying sample sizes (small, medium, large)
- Different noise levels (σ^2)
- Prior sensitivity in Bayesian models
- Misspecification of the outcome model

4.5 Expected Outcomes

- Improved accuracy and reduced bias in subgroup-specific treatment effect estimates
- Better uncertainty quantification via Bayesian credible intervals
- Enhanced interpretability of heterogeneous effects compared with non-Bayesian alternatives

5. EMPIRICAL APPLICATION

5.1 Dataset Description

A real-world healthcare dataset is employed to illustrate the proposed framework. The dataset includes:

- **Population:** Patients undergoing a specific treatment (e.g., diabetes management, cardiovascular therapy)
- **Sample size:** $N \approx 1,000 - 5,000$ patients
- **Covariates:** Demographics (age, sex), clinical measurements (blood pressure, HbA1c, cholesterol), comorbidities, lifestyle factors
- **Treatment variable:** Binary indicator of treatment assignment
- **Outcome:** Continuous (e.g., change in biomarker level) or binary (e.g., disease remission)

5.2 Preprocessing

- Missing values imputed using multiple imputation
- Continuous covariates standardized
- Categorical variables one-hot encoded
- Exploratory data analysis to check distribution and correlations

5.3 Subgroup Discovery Using MOB

- MOB is applied to detect covariates associated with heterogeneous treatment effects
- Splits are based on statistically significant parameter instability tests ($p < 0.05$)
- Terminal nodes define interpretable subgroups with distinct treatment-response patterns

5.4 Bayesian Flexible Modeling Within Subgroups

- Bayesian additive regression trees (BART) or hierarchical splines fitted within each subgroup
- Priors specified to allow partial pooling across subgroups
- Posterior distributions of treatment effects computed for subgroup-level CATE and individual-level ITE

5.5 Results

- **Subgroup characterization:** Example, older patients with high baseline risk responded differently than younger patients with low risk

- **Treatment effect estimates:** Posterior means and 95% credible intervals reported
- **Visualization:** Tree plots for subgroups, density plots for posterior treatment effects, and interaction maps for covariate-treatment relationships
- Comparison with standard MOB and BART applied to full dataset shows improved precision and interpretability

5.6 Interpretation

- The framework identifies clinically meaningful subgroups
- Bayesian modeling captures complex nonlinearities and quantifies uncertainty
- Personalized treatment recommendations can be derived from ITE distributions

6. DISCUSSION

6.1 Key Findings

- Integration of MOB and Bayesian flexible models provides a robust framework for identifying and estimating heterogeneous treatment effects
- The method successfully captures subgroup-specific nonlinear patterns and offers uncertainty quantification
- Simulation studies and empirical application demonstrate improved estimation accuracy, interpretability, and robustness compared to conventional methods

6.2 Methodological Advantages

- **Subgroup discovery:** MOB offers interpretable covariate-driven splits
- **Flexible modeling:** Bayesian models handle nonlinearities and high-dimensional covariates
- **Uncertainty quantification:** Posterior distributions allow credible intervals and probabilistic decision-making
- **Partial pooling:** Stabilizes estimates in small subgroups

6.3 Clinical and Policy Implications

- Supports precision medicine by identifying patients who are more likely to benefit from specific treatments

- Informs healthcare providers about potential risks and effectiveness across patient subgroups
- Can be extended to guide treatment guidelines and resource allocation in clinical settings

6.4 Limitations

- Computational complexity increases with large datasets and high-dimensional covariates
- Sequential integration of MOB and Bayesian modeling may miss subtle cross-subgroup interactions
- Requires careful prior specification in Bayesian models

6.5 Future Directions

- Joint modeling approaches integrating subgroup discovery and Bayesian inference
- Extension to longitudinal and time-to-event outcomes
- Application to adaptive clinical trials and dynamic treatment regimes
- Exploration of alternative Bayesian flexible models (e.g., Gaussian processes, Dirichlet process mixtures)

7. CONCLUSION

This study presents an integrated framework for **personalized treatment effect estimation** by combining **model-based recursive partitioning (MOB)** with **Bayesian flexible models**. The proposed approach effectively identifies subgroups with heterogeneous treatment effects and provides robust, interpretable estimates while quantifying uncertainty.

Key contributions include:

1. **Subgroup discovery:** MOB allows data-driven identification of covariate-defined patient subgroups exhibiting differential treatment responses.
2. **Flexible estimation:** Bayesian models capture nonlinear and complex relationships within subgroups, improving estimation accuracy and interpretability.
3. **Uncertainty quantification:** Posterior distributions provide credible intervals for both subgroup-level and individual-level treatment effects, supporting probabilistic clinical decision-making.

Simulation studies demonstrate that the integrated framework outperforms conventional MOB and standalone Bayesian models in terms of bias, RMSE, and coverage probability. An empirical application to healthcare data highlights its clinical relevance, showing meaningful subgroup stratification and individualized treatment effect estimation.

The framework offers a practical and principled solution for personalized medicine, enabling clinicians and policymakers to make informed treatment decisions. Future work may extend this approach to longitudinal data, dynamic treatment regimes, and joint modeling strategies to further enhance precision and generalizability.

8. REFERENCES

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